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From Hand-Drawn Schematics to **Structural Circuit** Netlists: A Deterministic Pipeline with Endpoint-Graph Wire Joining and a Human-Verified Connectivity Benchmark

BOSCO CHANAM¹ , CHRIS DCOSTA¹ , PRANAVESH KUMAR TALUPURI² , SHWETAMBARI CHIWHANE¹ , ASHAY KUMAR SINGH¹ , and ARGHADEEP DAS¹ 

¹Symbiosis Institute of Technology, Pune 412115, India (e-mail: boscochanam@gmail.com; chrisdcosta777@gmail.com; shwetambari.chiwhane@sitpune.edu.in; ashay.sit@gmail.com; arghadeep.das.work@gmail.com)

²Department of Computer Science, University of Southern California, Los Angeles, CA 90089 USA (e-mail: talupuri@usc.edu)

Corresponding author: Bosco Chanam (e-mail: boscochanam@gmail.com).

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ABSTRACT Recovering a netlist from a hand-drawn schematic requires deciding which wire strokes connect which component terminals. We present a deterministic pipeline for this structural connectivity task. An occlusion-first extractor masks components before binarization and returns wire segments with explicit endpoints. A typed endpoint graph links endpoints, pins and wire bodies under scale-relative tolerances, and a degree-budget completion step reconnects floating pins by bounded assignment. We evaluate on two real-image connectivity benchmarks from CGHD-1152: 31 images with human-verified nets, and 164 held-out photographs from 24 of the corpus's 25 drafters whose reference nets we derive from the dataset's own stroke and symbol annotations. With annotated component boxes, the join reaches component-pair micro-F1 0.890 on the human-verified set, on which its configuration was selected, and 0.711 on the held-out photographs. On both benchmarks it outperforms every deterministic baseline, including connected-component tracing and Hough line fitting, in paired tests with Holm correction. With annotated wires, its score stays within 0.005 of the native value when the images are rescaled from $0.5\times$ to $3\times$. Replacing the annotated boxes with a trained component detector gives micro-F1 0.627, and most of that loss comes from missed components. The wire extractor reaches F1 0.973 on 134 annotated images; on the held-out photographs, wire extraction is the main source of error. A vision-language model given the same component boxes reaches micro-F1 0.923 on the human-verified set: the pooled difference from our join is not significant, and per-image scores favor the model. We release the code, both benchmarks and the evaluation scripts.

INDEX TERMS circuit digitization, connectivity benchmark, graph-based connectivity, hand-drawn circuits, image-to-netlist, netlist extraction, schematic recognition, structural netlist, wire extraction, wire joining

I. INTRODUCTION

A circuit schematic encodes component types, component values and the connections between components in a drawing made for people. A netlist records the same connections in a form that software can check, simulate and search. Converting hand-drawn schematics into netlists would let course material, lab notebooks and legacy designs enter these tools without manual re-entry. This paper addresses the structural part of that conversion: given the components in an image, decide which component terminals share an electrical node.

We call this step *joining*.

Joining is hard for three reasons. First, wire endpoints extracted from binarized images are noisy: the same circuit drawn by different drafters yields endpoints at different offsets from the pins, and strokes stop short of or overshoot their targets. Second, component symbols overlap the wire endpoints that attach to them, so an endpoint near a component may belong to one of its pins or merely pass by. Third, T-junctions, collinear fragments and wire crossings all look alike to a proximity rule.

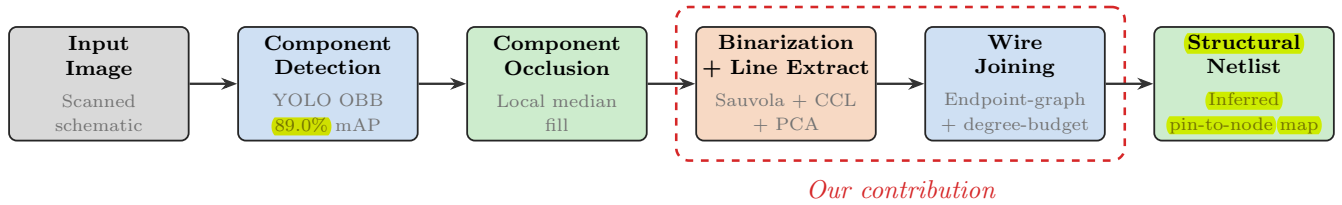


FIGURE 1. Pipeline for structural circuit-netlist recovery. The highlighted stages extract wire endpoints and join them into nets.

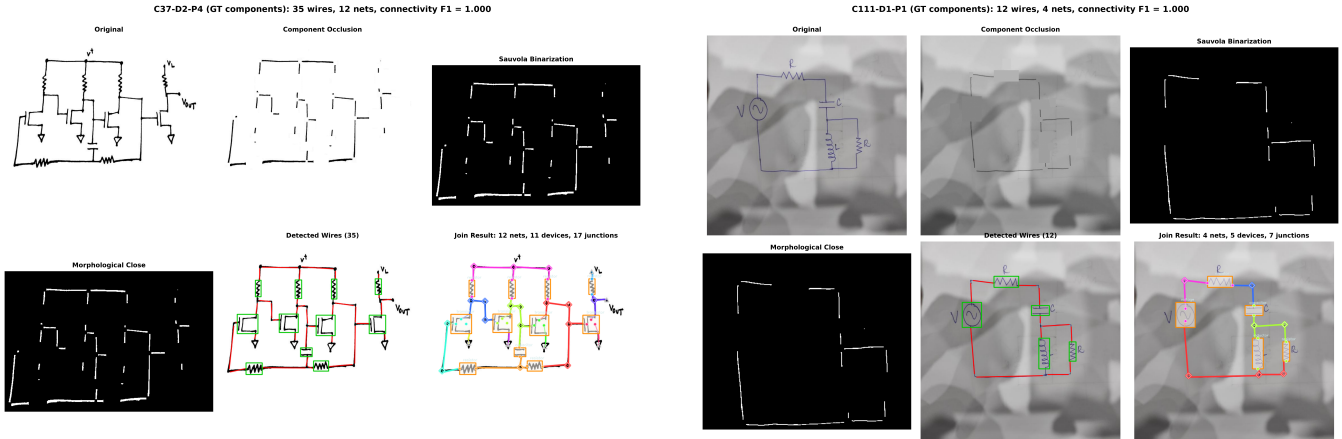


FIGURE 2. Pipeline stages on two hand-drawn circuits, joined with the scale-relative completion join (component-pair $F1 = 1.000$ on both). Left: a four-transistor analog cell (35 wires, 7 nets). Right: an RLC loop photographed on a textured surface (11 wires, 4 nets). Each panel shows the input, component occlusion, Sauvola binarization, morphological closing, detected wires (red) and the join result. In the join result each net has its own color; boxes are the annotated components, filled dots are device pins, diamonds are junctions and triangles are terminals and ground.

Most image-to-netlist systems erase the components and read nets from the connected components or contours of the remaining wire pixels [1], [2]. A single stroke gap then splits a net, and a single touching stroke merges two. Other systems pass the image to a large language model (LLM) and read connectivity from its answer [3]. We keep an explicit representation of every wire endpoint and decide each connection with geometric rules that can be inspected. This paper makes four contributions:

- 1) An *occlusion-first wire extractor* that fills each component region with its local median color before Sauvola binarization and returns line segments with PCA-estimated endpoints. It reaches wire-detection $F1$ 0.973 on 134 CGHD-1152 images (0.976 for the best of 36 variants).
- 2) A typed *endpoint-graph join* with five edge rules, component-first pin assignment and scale-relative tolerances, followed by *degree-budget completion*, which reconnects floating pins through a bounded-capacity assignment with a shared-component guard.
- 3) Two real-image connectivity benchmarks: 31 CGHD-1152 images with human-verified nets, and a 164-image held-out benchmark covering 24 of the 25 CGHD drafters whose reference nets are derived from the CGHD authors' own stroke-segmentation maps and symbol polygons. The derived reference agrees with the human nets on 17 overlapping images at micro- $F1$

0.939.

- 4) An evaluation protocol with paired significance tests under Holm correction, complexity and component-size strata, controlled image rescaling, end-to-end runs with a trained component detector, and a vision-language model (VLM) reference that receives the same component boxes as our join.

The output is a structural netlist; recognizing component values and device models is outside the scope of this work.

Section II reviews related work, Section III describes the method and Section IV the experimental setup. Section V reports results, and Section VI discusses them together with the scope and limitations of the study.

II. RELATED WORK

Several systems convert schematic images into machine-readable netlists. Kelly and Cole [4] presented a pattern-recognition workflow for printed schematics with an 86.4% success rate, combining optical character recognition (OCR) for component values with network-graph construction. SINA [1] pairs a YOLOv11 detector with connected-component labeling (CCL) and OCR, and uses a VLM only for reference-designator assignment, reporting 96.47% netlist-generation accuracy, a $2.72\times$ improvement over the prior state of the art. SINA derives nets from CCL proximity rather than from the VLM. Masala-CHAI [3] uses large language models to generate SPICE descriptions of analog

schematics at scale. The concurrent work of Peker et al. [2] builds an automated image-to-SPICE pipeline for printed and hand-drawn transistor-level circuits. It pairs YOLOv8/11 detection with a contour-based node-detection module, in which components are erased and nets are read from the contours of the residual wire pixels, and it validates netlists by LTspice simulation against a reference. On an in-house seven-topology MOSFET dataset it reports 93.33% (printed) and 85.33% (hand-drawn) whole-netlist success. Its node detection belongs to the connected-component family that we evaluate as a baseline on identical wire inputs (Section V-C); its dataset and metric differ from ours, so the headline numbers are not directly comparable. Bayer et al. [5] segment components and wires for graph extraction on the CGHD lineage. For hand-drawn inputs, object-detection pipelines [6] and the JUHCCR-v1 database [7] target component recognition on freehand sketches. Most of these pipelines build on general-purpose detectors such as the You Only Look Once (YOLO) family [8], [9] and OCR engines such as Tesseract [10], and target simulator input formats such as SPICE [11]. SPICE-level modeling also supports hardware work in neighboring domains, including memristor-based neuromorphic circuits that realize brain-inspired learning [12] and memristive decision-making circuits for autonomous navigation [13]. We focus on the joining stage and evaluate it with both annotated and detected components.

Progress depends on annotated corpora. The CGHD dataset [14] provides component- and wire-level annotations for handwritten schematics. In the analog/mixed-signal (AMS) domain, AMSnet 2.0 [15] contributes a large database built with AI-assisted segmentation for net detection, and AMSBench [16] evaluates multimodal LLMs across AMS circuit tasks. DiagramNet [17] introduces a framework and dataset for non-standard system-level diagrams, including a connection-recognition task scored by F1. To our knowledge, no public benchmark reports net-level connectivity on hand-drawn circuit images; Peker et al. [2] report whole-netlist success on an in-house hand-drawn set.

General-purpose VLMs and LLMs have been applied broadly to electronic design [18] and to circuit understanding. Kulkarni et al. [19] proposed Near Sight Correction with VLMs, reporting 87.4% accuracy on circuit visual question answering and 0.87 F1 on connection-matrix prediction. On the DiagramNet connection task [17], general models score low F1 (GPT-5 0.029, Claude-Sonnet-4 0.265, Gemini-2.5-Pro 0.008), whereas a task-tuned 3B model reaches 0.735. AMSBench [16] reports a similar gap for AMS connectivity. These results vary with task and input, which motivates measuring VLM connectivity directly on our benchmark (Section V-I).

Classical wire extraction relies on edge detection, morphological operations and the Hough transform [20], usually after adaptive binarization such as Sauvola thresholding [21] and connected-component labeling [22]. The legacy join in our pipeline uses radius-based proximity: each wire endpoint takes every component pin within a fixed threshold (30 px),

and connections are merged transitively with union-find [23]. When several components fall within the threshold, this rule assigns both pins of a two-terminal part to the same net. We model joining instead as a typed endpoint graph that uses component geometry, and we resolve residual ambiguity with degree-budget completion, a bounded b -matching formulation [24], [25] that caps the number of recovery connections per pin. Graph neural networks for netlist reverse engineering [26] learn representations over already-extracted gate-level graphs; our graph is geometric, and its edges encode spatial relations between endpoints and pins. Netlistify [27] predicts terminal connectivity with a Transformer and reports a 12.4% F1 gain over AMSnet on printed AMS schematics; such learned models need connectivity training data, whereas our join needs none. Open hand-drawn efforts such as CircuitNet [28] combine traditional image processing with a classifier and a heuristic netlist generator.

III. METHOD

A. PIPELINE OVERVIEW

Figure 1 summarizes the six stages: (1) component detection with an oriented-bounding-box (OBB) YOLO model; (2) component occlusion, which fills each component region, enlarged by 15% of its box size (at least 5 px), with the median color of its surroundings; (3) Sauvola binarization ($k = 0.285$, window 67 px) inside the union of component boxes plus 10 px padding; (4) connected-component labeling after a 3×3 elliptical closing, discarding components under 28 px; (5) wire-segment and endpoint estimation by principal component analysis (PCA) of each connected component, followed by overlap deduplication (12° , 8 px) and an anchor filter with a 16 px endpoint distance; and (6) joining with the endpoint graph and degree-budget completion. The output maps each component pin to an electrical node. Figure 2 shows two circuits at each stage.

B. ENDPOINT-GRAPH JOIN MODEL

Given wire segments $\mathcal{W} = \{w_1, \dots, w_m\}$ with endpoints $\text{ep}(w_i) = (a_i, b_i)$ and component pins $\mathcal{P} = \{p_1, \dots, p_n\}$, we construct an undirected graph $G = (V, E)$ with $V = \text{eps}(\mathcal{W}) \cup \mathcal{P}$ and five edge types.

Edge type 1 (wire body): For each wire w_i , add edge (a_i, b_i) so that both endpoints belong to the same net.

Edge type 2 (endpoint-endpoint): For endpoints e_i, e_j on different wires with $\|e_i - e_j\| \leq \tau_{\text{join}}$, add an edge. This rule connects fragments, corners and junctions.

Edge type 3 (endpoint-pin): Each endpoint connects to at most one pin. The component-first assignment described below chooses the pin. If no component is assigned, the endpoint takes the pin within $d \leq \tau_{\text{pin}}$ with the smallest score, where the score is $d[1 - \alpha \max(0, \cos \theta)]$ with $\alpha = 0.35$ and θ the angle between the outward wire direction and the endpoint-to-pin vector (for $d \leq 10^{-6}$ px the score is d). If neither step finds a pin, no edge is added.

Edge type 4 (endpoint-wire-body): For an endpoint e and a different segment w_j , add an edge if the point-to-segment

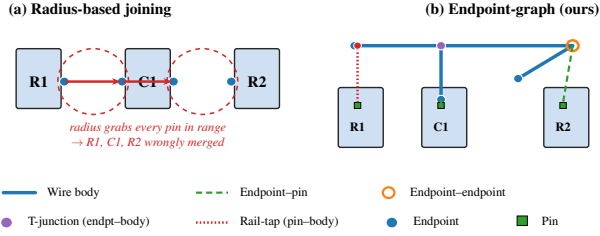


FIGURE 3. Joining approaches. (a) Radius-based join (legacy): each wire endpoint takes all pins within a fixed threshold, which over-merges. (b) Endpoint-graph model: typed edges and component-first pin assignment; directional scoring applies only to the fallback pin search.

distance is at most τ_i and the normalized projection parameter lies in $[0.15, 0.85]$. This rule captures T-junctions away from segment ends.

Edge type 5 (pin-wire-body): For a pin p and a segment w_j , add an edge if the point-to-segment distance is at most τ_i and the projection parameter lies in $[0.05, 0.95]$. This rule captures pins tapped onto a rail.

The tolerances are scale-relative with fixed pixel clamps: $\tau_{\text{pin}} = \min(60, \max(24, 0.62s))$, $\tau_{\text{join}} = \min(28, \max(11, 0.30s))$ and $\tau_i = \min(20, \max(8, 0.20s))$, in pixels. Here s is the median component-box diagonal of the image (the upper middle value for an even count), with a wire-length scale as fallback when component geometry is unavailable. The multipliers and clamps are set by hand. Because s is one scalar per image, these three tolerances do not vary with component size inside a drawing.

Component-first pin assignment: For edge type 3, each endpoint is first assigned to a component. Its distance to a component is 0 inside the component's OBB and otherwise the distance to the nearest OBB edge (axis-aligned box distance when OBB vertices are unavailable). The endpoint is assigned to the nearest component within $\max(\tau_{\text{pin}}, 0.5\delta_c)$, where δ_c is that component's own box diagonal. Because this radius grows with each component's diagonal, endpoint-to-component binding follows the size of each component in addition to the image-level scale s . Within the assigned component, the endpoint takes the nearest pin. Figure 3 contrasts the legacy radius join with the endpoint graph.

Nets are the connected components of G , accumulated with union-find and projected onto pins. The result maps each pin to a net identifier.

C. DEGREE-BUDGET COMPLETION

After base joining, some pins lie in a net that contains only pins of their own component. Completion treats these *floating* pins as candidates for recovery, excluding inert types (ground, supply markers, text, antennas, probes, junctions and terminals).

Completion keeps the base nets and proposes connections from each floating pin p to pins q of other components in other nets. A candidate's distance must not exceed the reach $r = \rho\tau_c$, where $\tau_c = \min(60, \max(24, 0.62s))$ px and $\rho = 4$. For each candidate we look for a wire witness: a detected segment

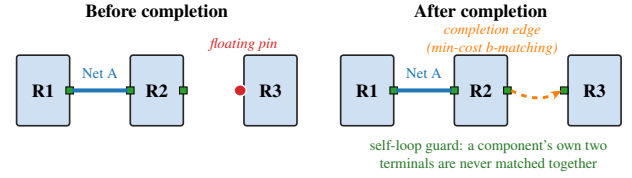


FIGURE 4. Degree-budget completion. Left: after base joining, some pins are floating (connected only to their own component). Right: completion reconnects floating pins by min-cost b -matching (orange dashed); a guard rejects merges between nets that already share a component.

whose endpoints lie near p and q in either orientation. The witness cost is the smallest, over segments and orientations, of the larger endpoint-to-pin distance, bounded by r . Without a witness, the candidate cost is $\|p - q\|$.

The candidates form a rectangular assignment matrix with one row per floating pin. Each target pin supplies up to three slot columns, limited by its candidate demand, and each row has a private dummy column of cost $1.5r$ for opting out. Unavailable assignments cost 10^6 . The Hungarian algorithm [24] (`scipy.optimize.linear_sum_assignment`) assigns every row to one column, with at most one row per column; the target slots implement bounded-capacity matching [25].

Matches are applied in sequence. Dummy and unavailable assignments add no edge. A merge is skipped if both pins are already in the same net or if their nets already share a component; this guard acts on completion merges only. Completion edges have no corresponding wire segment, much like a manual pin merge in a schematic editor. Figure 4 illustrates the step.

IV. EXPERIMENTAL SETUP

A. DATASETS AND IMAGE PROVENANCE

All real images come from CGHD-1152 [14], a public corpus of hand-drawn circuit images organized by drafter. We use four evaluation sets. The *wire benchmark* has 134 images with human wire labels and annotated component boxes. The *human-verified net benchmark* has 31 images from 10 drafters. The 34 annotated images were chosen by circuit content, not appearance: ten initial images spanning a range of complexity, and the rest sampled at even intervals of electrical-component count from the circuits whose parts all belong to the scored device types. Image background was not a selection criterion, but the selected copies have mostly plain backgrounds, partly because 16 of them are stroke maps (see below). Three annotated images were excluded before scoring (one unverified, two failing physical-sanity checks). Net proposals were bootstrapped by a conservative endpoint-graph join over the annotated wires and corrected by one annotator in a purpose-built verification tool. The 31 images contain 3–14 electrical components each (median 7); 15 have at most five and 12 have at least ten. The *independent held-out benchmark* (164 images, 24 drafters) and the *synthetic suite*

(15 authored circuits) are described in Sections V-D and V-A.

The 134 wire-benchmark and 31 net-benchmark images are 704×704 copies from a Roboflow export of CGHD. The export resized each original without preserving its aspect ratio. Of the 31 net-benchmark images, 14 are also flipped or rotated relative to the CGHD file, and 16 (45 of the 134 wire-benchmark images) are CGHD's binary stroke-segmentation renderings instead of photographs. Wire and net labels were drawn on these copies. The held-out benchmark uses the original CGHD photographs.

B. CONNECTIVITY METRIC

For each image we form the set of unordered pairs of electrical components that share at least one net, and compare predicted with reference pairs. The electrical subset contains resistors, capacitors, inductors, diodes, transistors, sources and integrated circuits; junctions, terminals, ground symbols and crossovers affect the score only through the connections they mediate. The metric ignores which terminal of a component carries a connection. Micro-F1 pools true positives (TP), false positives (FP) and false negatives (FN) over images, so a dense board contributes in proportion to its connections; it is our primary metric. Macro-F1, the mean of per-image F1, is reported alongside.

C. STATISTICAL PROTOCOL

All comparisons are paired over images. We report 95% percentile confidence intervals (CIs) from an image-level bootstrap with $B = 10,000$ resamples, and test each difference with three procedures: the bootstrap itself, a sign-flip permutation test that swaps the two methods' outputs within each image (exact when at most 22 images differ, otherwise 100,000 Monte Carlo draws), and a Wilcoxon signed-rank test on per-image F1. The first two test the micro-F1 difference; the Wilcoxon test compares images. Holm's correction is applied within each procedure across the comparisons of a table. We also count per-image wins, ties and losses (W/T/L). Unless stated otherwise (Section V-G), every join receives annotated component boxes and the pipeline's own detected wires.

V. RESULTS

A. SYNTHETIC EVALUATION

We authored 15 circuits with 3–6 components and 2–6 nets, including parallel resistors, series dividers, Wheatstone bridges, RC/RL circuits, diode loops and a 6-component ring. Each circuit has a known netlist, component values and a test voltage source. We inject five error types at four severity levels (L1–L4) on top of the clean level L0: Gaussian endpoint jitter (σ from 3 px at L1 to 14 px at L4); cut-short wires, shortened inward by 4 px at L1 to 22 px at L4; wire drops with probability up to 0.20 at L4; wrong-pin snaps with probability from 0.02 at L2 to 0.10 at L4; and anchor displacement, which moves one endpoint by up to 30 px at L1 and 80 px at L4. Besides component-pair F1, we report simulation accuracy: the percentage of seeds for which every source current of the

TABLE 1. Synthetic join leaderboard: mean component-pair F1 over 15 circuits $\times$ 8 seeds at each error level.

Strategy	L0	L1	L2	L3	L4
Scale-rel. graph + completion (ours)	1.00	1.00	1.00	0.96	0.95
Rescue graph + completion	1.00	1.00	1.00	0.95	0.94
Rescue graph (base)	1.00	1.00	1.00	0.94	0.90
Scale-rel. graph (base)	1.00	1.00	1.00	0.94	0.85
Radius union-find (legacy)	1.00	0.97	0.90	0.56	0.36
Nearest-pin + bbox anchor	1.00	0.97	0.89	0.38	0.32

TABLE 2. Scale-relative completion join on eight of the 15 synthetic circuits at L4 (16 seeds).

Circuit	Comps	F1	Prec	Sim. acc.
Parallel R–R	3	1.00	1.00	25%
R–R divider	3	0.99	1.00	69%
RL series	3	0.99	1.00	81%
Diode–R loop	3	0.99	1.00	69%
Dual R loops	4	0.93	0.90	12%
4-component loop	4	0.94	0.91	56%
Wheatstone bridge	6	0.82	0.85	6%
6-component ring	6	0.95	0.92	50%

predicted netlist, simulated with the authored values, matches the clean reference within 5%.

Table 1 and Figure 5 compare the join strategies. At L4, the scale-relative graph with completion scores F1 0.95, the rescue graph with completion 0.94, the rescue graph alone 0.90, the scale-relative graph alone 0.85 and radius union-find 0.36. The rescue graph differs from the scale-relative graph by a 12 px end-extension and a dead-end rescue step that gives dangling wire ends a longer directional reach. Table 2 breaks down our join by circuit at L4. Precision is at least 0.85 on every circuit. The Wheatstone bridge is the hardest case, with F1 0.82 and simulation accuracy 6%; the parallel R–R circuit keeps F1 1.00 at 25% simulation accuracy, so the two measures capture different properties.

B. WIRE DETECTION

Table 3 and Figure 6 summarize the best of 36 extractor variants on the 134-image wire benchmark. Sauvola binarization with a 16 px anchor filter reaches F1 0.976 (median per-image F1 1.000; 87% of images at or above 0.90). Otsu thresholding reaches 0.789 and adaptive Gaussian thresholding 0.845. The configuration in the table deduplicates overlapping segments at $10^\circ/18$ px; the deployed $12^\circ/8$ px setting scores 0.973 with the same anchor filter. The score does not depend on image modality: F1 is 0.967 on the 89 photographs and 0.986 on the 45 stroke maps.

C. HUMAN-VERIFIED BENCHMARK

Table 4 reports every join configuration and baseline on the 31-image human-verified benchmark, together with the paired tests, and Fig. 7 plots the micro-F1 values with their CIs. Our join, degree-budget completion on the scale-relative base, reaches micro-F1 0.890 (P 0.919, R 0.864; macro-F1 0.901; TP/FP/FN 418/37/66; 95% CI [0.855, 0.924]).

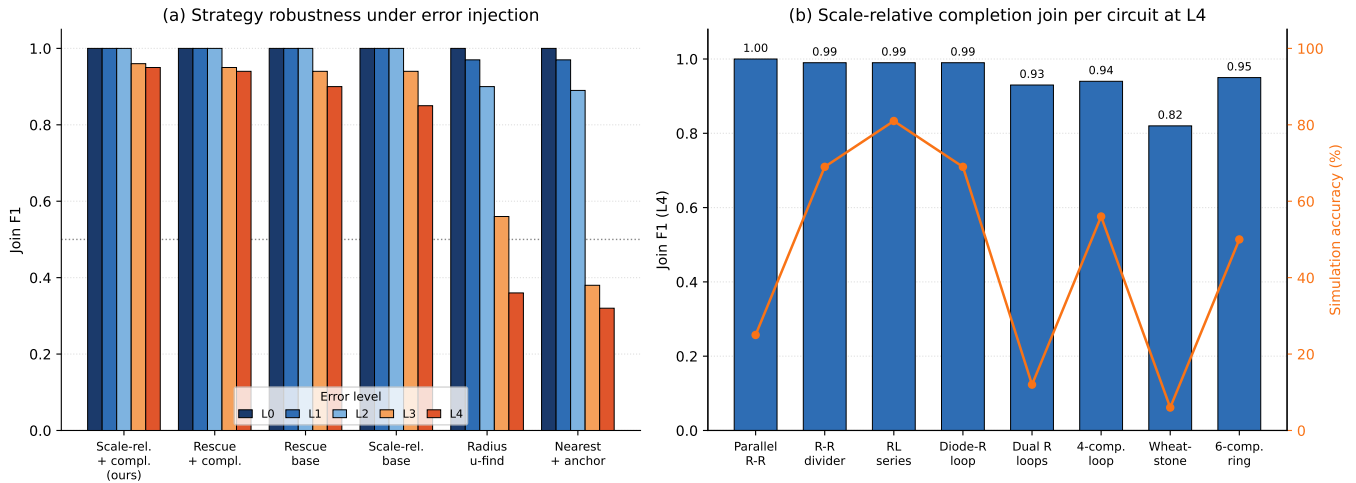


FIGURE 5. (a) Join strategies under error injection (15 circuits $\times$ 8 seeds): mean component-pair F1 at each severity level (L0–L4, dark to light to red). (b) The scale-relative completion join per circuit at L4 (16 seeds): bars are component-pair F1, the orange line is simulation accuracy.

TABLE 3. Wire detection on 134 images with annotated component boxes. Sauvola rows use $10^\circ/18$ px overlap deduplication.

Configuration	F1	Prec	Rec
Sauvola + anchor 16 px (ours)	0.976	0.973	0.978
Sauvola + anchor 12 px	0.973	0.974	0.972
Sauvola ($w=61$) + anchor 12 px	0.959	0.944	0.974
Sauvola ($k=0.2875$) + anchor 14 px	0.950	0.921	0.980
Otsu threshold + CCL	0.789	0.796	0.783

Sauvola, anchor 16px (ours)	0.976
Sauvola, anchor 12px	0.973
Sauvola ($w=61$)	0.959
Sauvola ($k=0.2875$)	0.950
Sauvola (alt.)	0.949
adaptive Gaussian (skeleton)	0.845
Otsu	0.789
Triangle (skeleton)	0.758

FIGURE 6. Wire detection F1 across thresholding methods and pipeline variants (134 images, annotated component boxes). The adaptive Gaussian and Triangle bars use skeleton extraction (F1 = 0.845 and 0.758).

It beats every deterministic comparator on every test after Holm correction. The margin over the prior default, which applies the same completion to the rescue base, is $+0.061$ [$+0.025, +0.100$] with 11 wins, 18 ties and 2 losses. Completion lifts the scale-relative base from 0.816 to 0.890, mostly through recall (0.727 to 0.864). The scale-relative base is also more precise than the rescue base (0.929 versus 0.811), and completion keeps most of that precision (0.919, against 0.789 for completion on the rescue base). Leaving out any one image moves our micro-F1 within 0.885–0.900, and no

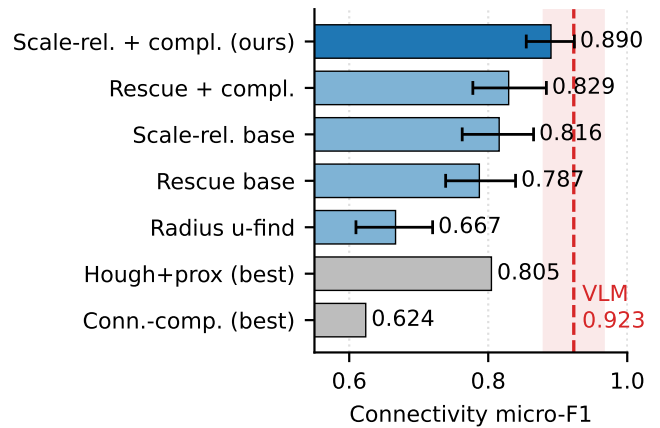


FIGURE 7. Micro-averaged component-pair F1 on the 31-image human-verified benchmark with annotated component boxes and detected wires. Error bars are 95% bootstrap CIs; the dashed line marks the VLM reference (Claude Opus 4.8).

deletion changes the sign of any comparison. A sweep of the completion reach over $\rho \in [3, 5]$ gives macro-F1 0.895–0.903.

Several image-to-netlist systems erase component regions and read nets from the remaining wire pixels with connected components or contours [1], [2]. On our detected wires, the best connected-component configuration reaches micro-F1 0.624 at precision 0.965: it rarely merges wrongly, but every stroke gap splits a net (recall 0.461). Hough line fitting [20] with proximity linking reaches 0.805 at its best linking tolerance. Both baselines use the best configuration of their sweep on these 31 images, which favors them.

We also tried to run five public image-to-netlist systems on the benchmark. The printed-schematic interpreter of Kelly and Cole [4] runs, but on hand-drawn scans it over-segments wires into spurious components and recovers almost no connectivity. Netlistify [27] reproduces on its own printed AMS test data, but its released model lacks the diode, inductor and

TABLE 4. Component-pair connectivity on the 31-image human-verified benchmark (annotated component boxes; detected wires unless noted). F1, P and R are micro-averaged. * is the micro-F1 of ours minus the comparator with 95% paired bootstrap CI; Holm-adjusted p values are given for the bootstrap, the permutation test on micro-F1 and the Wilcoxon test on per-image F1; W/T/L counts per-image wins, ties and losses from our side.

Method	F1	P	R	F1 _{macro}	Δ [95% CI]	Holm p (boot / perm / Wilcoxon)	W/T/L
<i>Endpoint-graph joins</i>							
Scale-rel. graph + completion (ours)	0.890	0.919	0.864	0.901	—	—	—
Rescue graph + completion	0.829	0.789	0.874	0.853	+0.061 [+0.025, +0.100]	<0.001 / 0.001 / 0.009	11/18/2
Scale-rel. graph (base)	0.816	0.929	0.727	0.838	+0.075 [+0.038, +0.113]	<0.001 / 0.002 / 0.005	14/16/1
Rescue graph (base)	0.787	0.811	0.764	0.813	+0.103 [+0.065, +0.142]	<0.001 / <0.001 / 0.002	17/14/0
<i>Baselines on the same detected wires</i>							
Radius union-find (legacy)	0.667	0.698	0.638	0.674	+0.224 [+0.166, +0.281]	<0.001 / <0.001 / <0.001	25/6/0
Hough + proximity (best)	0.805	0.750	0.868	0.847	+0.086 [+0.045, +0.127]	<0.001 / 0.002 / 0.030	19/8/4
Connected components (best)	0.624	0.965	0.461	0.611	+0.267 [+0.136, +0.409]	<0.001 / <0.001 / <0.001	21/8/2
<i>References</i>							
Ours with annotated wires	0.890	0.913	0.868	0.916	—	—	—
VLM (Claude Opus 4.8)	0.923	0.970	0.880	0.949	−0.033 [−0.078, +0.008]	0.12 / 0.15 / 0.030	7/11/13

IC classes in our circuits, and its connectivity model is trained on printed thin-line drawings. Masala-CHAI [3] delegates connectivity to a proprietary LLM behind a paid API, which our VLM reference (Section V-I) covers. The code archive of SINA [1] could not be extracted, and CircuitNet [28] ships notebooks without trained weights. The CCL and Hough baselines therefore stand in for these approaches on identical inputs.

Replacing the detected wires with the annotated wires, with boxes held fixed, changes micro-F1 from 0.8903 to 0.8898 and macro-F1 from 0.901 to 0.916. On this benchmark, the extracted wires are as good as the annotated ones for joining. The join also scores alike on both image modalities: micro-F1 is 0.883 on the 15 photographs and 0.895 on the 16 stroke maps.

D. INDEPENDENT HELD-OUT BENCHMARK

The 31-image benchmark has three weaknesses: it is small, its labels were bootstrapped from a graph join similar to ours, and our configuration was chosen on it. We therefore built a second benchmark whose labels come from the CGHD authors' own annotations and whose images were not used to develop or tune the join. The extractor configuration was selected earlier on the wire benchmark, whose resized copies include 25 of the held-out drawings; excluding those 25 changes none of the results below.

Reference construction. CGHD provides binary stroke-segmentation maps and symbol instance polygons for a subset of its images. Extending the dataset's own wire-generation tooling, we subtract the dilated symbol polygons from the stroke map and take the remaining connected components as conductors. A conductor connects every polygon it touches, except text; junction polygons merge the conductors they touch. At a crossover polygon, the arms are paired straight through by their local direction ($180^\circ \pm 45^\circ$), and a crossover that cannot be paired is flagged as ambiguous. Composite outlines (transformers, optocouplers and relays that contain sub-parts) group their parts but take no contacts. Of 257 images, 179 pass all quality checks; the others have an ambiguous

TABLE 5. Independent held-out benchmark (164 CGHD photographs, 24 drafters, 3874 reference pairs; annotated component boxes). * is ours minus the comparator in micro-F1, with 95% paired bootstrap CI. Every difference has Holm-adjusted $p = 0.0006$, the resolution floor of the tests.

Method	F1	P	R	Δ [95% CI]	W/T/L
Ours	0.711	0.766	0.664	—	—
Rescue + compl.	0.659	0.622	0.699	+0.053 [0.033, 0.074]	85/44/35
Scale-rel. base	0.595	0.785	0.479	+0.117 [0.097, 0.138]	112/46/6
Rescue base	0.592	0.631	0.558	+0.119 [0.097, 0.142]	113/31/20
Radius u-f	0.518	0.478	0.565	+0.194 [0.159, 0.229]	136/13/15
Hough	0.480	0.328	0.895	+0.231 [0.168, 0.282]	90/22/52
Conn. comp.	0.594	0.798	0.473	+0.117 [0.083, 0.156]	117/12/35

crossover (29), an isolated part (17) or an over-connected part (28).

Validation. On the 17 images shared with the human-verified benchmark, the reference agrees with the human nets at micro-F1 0.939 (P 0.993, R 0.891; exact on 11 of 17; 0.964 on the 15 clean images). We inspected every disagreement. Most are pairs the reference misses because of stroke gaps or ambiguous crossings, and seven follow from a convention: the human nets bridge switches. In five pairs, the reference was right and the human labels wrong. The reference mainly under-connects, so it understates the precision of every method. We fixed the quality filter and a junction-handling rule after inspecting these 17 overlaps.

Protocol. The held-out set contains the 164 clean images outside the 31, with 3874 reference pairs from 24 drafters (all except drafter_0, which the detector's training also excluded). The input is the CGHD photograph with its aspect ratio preserved and its long side resized to 1024 px. We chose this resolution on wire-level labels of the 52 wire-benchmark overlaps, before computing any join score. Unlike the 31-image benchmark, which kept only circuits whose parts all belong to the scored device types, the held-out set also contains switches, potentiometers, logic gates, speakers, thyristors and transformers (94 of the 164 images contain at least one); these parts take part in the topology but are not scored.

Results. Table 5 gives the results. Our join reaches micro-F1 0.711 [0.671, 0.748] (P 0.766, R 0.664; macro-F1 0.724) and beats every comparator, with the same ordering of the endpoint-graph variants as on the human-verified set. Hough linking keeps recall high (0.895) but merges many nets (precision 0.328); the best Hough configuration tuned on this set itself reaches 0.509. The result holds under three changes to the evaluation set: excluding every other photograph of the 31 benchmark drawings (155 images) gives 0.707, and scoring all 257 images including the unclear ones gives 0.626. Across the 24 drafters, our micro-F1 ranges from 0.465 to 0.917 (median 0.709, interquartile range 0.658–0.770), and our join is the best of the evaluated methods for 15 drafters.

Scoring conventions. The primary scoring follows the human-verified benchmark: it scores only the electrical device types, treats switches as components that are not scored, and lets separate ground symbols join nodes only through drawn wires. We rescored the same predictions under three alternatives, applied symmetrically to reference and predictions. Adding switches, potentiometers, photoresistors, fuses, lamps, varistors, crystals, speakers and microphones to the scored set (577 more reference pairs in 76 images) gives 0.709, with a margin of +0.055 [+0.036, +0.077] over the rescue-graph completion. Treating every switch as closed gives 0.712 (+0.052). Merging all ground symbols into one node, and all negative-supply symbols into another, adds 778 pairs in 43 images and gives 0.679 (+0.035 [+0.012, +0.059]); the merged ground net concentrates many pairs, so one missed ground attachment costs several. With all three changes combined, our join scores 0.676 and leads every baseline, with Holm-adjusted $p \leq 0.009$.

A human audit of 40 randomly sampled held-out images [RESULT PENDING].

Gap to the human-verified benchmark. The drop from 0.890 to 0.711 comes mainly from the input images. The held-out photographs belong to the CGHD subset for which the dataset provides stroke maps, and for this subset the 704×704 benchmark copies are mostly those stroke maps (16 of the 17 overlaps). Many of these photographs show grid paper or cluttered backgrounds, which the 31-image copies rarely contain; on them, wire F1 falls to about 0.80–0.82, against 0.98 on the benchmark copies. On the 17 overlapping images, our join scores 0.896 on the benchmark copies and 0.728 on the photographs at long side 1024. Re-weighting the held-out images to the circuit-size distribution of the 31 raises our score by about 0.01, and scoring against the human nets instead of the reference changes it by about 0.01. Input quality explains much of the rest, but not all of it: with the CGHD stroke maps as input instead of the photographs, our held-out score rises only to 0.775. The residual is consistent with some optimism in the 31-image score, on which our configuration was selected. Across both benchmarks, our join ranks first and the endpoint-graph variants keep their order; the classical baselines do not keep theirs (Hough ranks below connected components on the held-out set).

TABLE 6. Micro-F1 by number of electrical components per image. The VLM was run on the human-verified benchmark only.

Components	Images	Ours	Rescue + compl.	Hough	VLM
<i>Human-verified benchmark (31 images)</i>					
≤5	15	0.940	0.933	0.892	0.982
6–9	4	0.846	0.817	0.843	0.902
≥10	12	0.886	0.808	0.779	0.912
<i>Held-out benchmark (164 images)</i>					
≤5	41	0.845	0.856	0.835	–
6–9	41	0.779	0.756	0.701	–
10–19	60	0.694	0.656	0.593	–
≥20	22	0.693	0.608	0.347	–

E. ROBUSTNESS

1) Circuit complexity

Table 6 splits both benchmarks by the number of electrical components. On the 31 images, F1 falls with complexity for every method, including the VLM: across images, our F1 correlates with component count at Spearman $\rho = -0.53$ ($p = 0.002$). Our recall drop from the ≤5 to the ≥10 stratum, -0.068 [-0.18 , $+0.05$] (permutation $p = 0.32$), is not significant. On images with at least ten components, our join beats the rescue-graph completion, the scale-relative base and Hough by +0.078 to +0.106, with every CI excluding zero, and trails the VLM by -0.026 [-0.087 , $+0.030$]. On the held-out benchmark, micro-F1 falls from 0.845 for at most five components to 0.693 for 20 or more, while Hough falls to 0.347.

2) Component-size variation within an image

A single scale s per image could fail when component sizes vary within a drawing. On the 31 images, the overall dispersion of component sizes is unrelated to F1 ($\rho = +0.18$, $p = 0.33$). The ratio of the largest to the smallest electrical-component diagonal correlates with F1 at $\rho = -0.60$, but it also grows with component count ($\rho = 0.62$); controlling for count, the partial correlation is -0.41 ($p = 0.024$) for our join and -0.34 for the VLM. The size of the largest component relative to s has partial correlation near zero with F1. The three images that combine an IC with small discrete parts score 0.854 against 0.898 on the other 28.

3) Controlled rescaling

We resampled all 31 images by factors from 0.35 to 3 and re-projected the annotated boxes and wires (Fig. 8). With annotated wires, our join stays within ± 0.005 of its native score from $0.5\times$ to $3\times$, with every CI including zero. It drops only at $0.35\times$ (-0.035 [-0.061 , -0.005]), where the lower pixel clamps bind for every image. With annotated wires, radius union-find with its fixed 30 px radius falls from 0.690 at native scale to 0.429 at $2\times$ and 0.153 at $3\times$. With wires re-extracted from each rescaled image, the full pipeline scores 0.580, 0.690, 0.811, 0.890, 0.877, 0.866 and 0.866 at 0.35, 0.5, 0.75, 1, 1.5, 2 and $3\times$. Upscaling costs at most 0.024 (CIs include zero). The loss under downscaling comes from

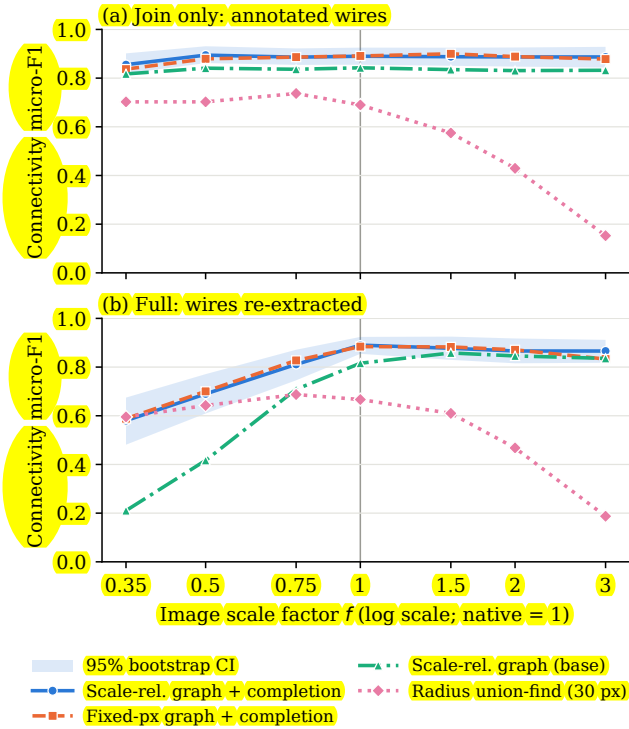


FIGURE 8. Component-pair micro-F1 on the 31 images resampled by factor f (native $f = 1$). (a) Join only, with annotated wires rescaled with the image. (b) Full pipeline, with wires re-extracted from each rescaled image. Components are annotated boxes in both panels; shaded bands are 95% bootstrap CIs for our join.

wire extraction, whose Sauvola window, minimum component area and occlusion margins are set in pixels: at $0.35 \times$ the extractor finds 241 of 675 annotated segments. Completion recovers much of that loss; the scale-relative base alone scores 0.21 at $0.35 \times$.

To separate the contribution of the scale-relative tolerances, we ran the same algorithm with fixed pixel tolerances (τ_{pin} , τ_{join} , τ_t of 30, 14 and 10 px, completion reach 120 px). It stays within 0.016 of our join between $0.5 \times$ and $2 \times$. The scale-relative version wins only where the fixed values are most mismatched: by $+0.015$ with annotated wires at $0.5 \times$ and by $+0.033$ in the full pipeline at $3 \times$, both with CIs excluding zero. Most of the scale robustness therefore comes from component-relative endpoint assignment and completion; the scalar tolerances $\tau = k \cdot s$ contribute a smaller share. The clamps are tuned to native resolution: $k \cdot s$ falls outside a clamp for 39% of images at native scale, and removing the clamps costs 0.023 in the full pipeline at native scale.

4) Synthetic mixed-size circuits

We redrew one component per synthetic circuit at 0.5 , 2 or 3 times its size, keeping only geometrically clean variants, and applied the L1–L4 error model. Enlarging one component costs our join about 0.005 in mean F1, and it stays about 0.03 above the fixed-pixel variant. Shrinking one component costs 0.012. The failure case is a Wheatstone bridge with one

TABLE 7. Edge-rule ablation on the 31-image benchmark, for the base graph and the full pipeline with completion.

Configuration	Base graph		Full pipeline	
	F1	TP/FP/FN	F1	TP/FP/FN
All edge types (ours)	0.816	352/27/132	0.890	418/37/66
No T-junctions	0.816	352/27/132	0.890	418/37/66
No rail taps	0.816	352/27/132	0.890	418/37/66
No T-junctions, no rail taps	0.816	352/27/132	0.890	418/37/66
No directional preference	0.816	352/27/132	0.890	418/37/66
Fixed-pixel base tolerances	0.820	356/28/128	0.890	418/37/66

TABLE 8. Mechanism ablations on the 31-image benchmark. “Changed” counts images whose predicted pairs differ from the reference arm.

Arm	TP/FP/FN	F1	F1 _{macro}	Changed
Occlusion on (ours)	418/37/66	0.890	0.901	–
Occlusion off	103/26/381	0.336	0.294	31/31
Guard on (ours)	418/37/66	0.890	0.901	–
Guard off	418/37/66	0.890	0.901	0/31
Witness required	413/37/71	0.884	0.907	6/31

40×15 px arm; in all five variants, the scale-relative methods and radius union-find add one spurious pair even on clean wires, while the fixed-pixel graphs stay exact. A component much smaller than the median is thus the weak point of a median-derived tolerance.

F. ABLATIONS

Table 7 removes individual edge rules from the base graph, with and without completion. Disabling T-junctions, rail taps or directional preference leaves the aggregate base-graph counts unchanged, so these rules give no measurable gain on this benchmark. Fixed-pixel base tolerances raise base micro-F1 from 0.816 to 0.820. With completion, every row ties at TP/FP/FN 418/37/66. The rescaling and mixed-size experiments (Section V-E) test the tolerance design under conditions that this benchmark does not cover.

Table 8 tests the remaining mechanisms. Removing only the occlusion step, with cropping and all extractor settings held fixed, exposes component pixels to the extractor, which then returns 466 wire segments instead of 666; micro-F1 falls from 0.890 to 0.336 (P 0.798, R 0.213), and predictions change on all 31 images. The shared-component guard fires on 18 of 31 images and rejects 38 candidate merges. These merges would join two pins of the same component, which the component-pair metric cannot see, so the scores are unchanged. Requiring a wire witness for every completion edge changes six images and lowers micro-F1 to 0.884 while raising macro-F1 to 0.907. The rescaling experiment adds two further arms, fixed-pixel tolerances and unclamped tolerances (Section V-E), and Section V-G decomposes the loss from detected components.

TABLE 9. End-to-end joining with detected components on the 31-image benchmark. The lower block replaces annotated boxes with detector output one factor at a time (confidence 0.5); Δ is the change from the previous row.

Components	F1	Prec	Rec	TP/FP/FN
Annotated boxes	0.890	0.919	0.864	418/37/66
Detector, conf 0.25	0.613	0.575	0.655	317/234/167
Detector, conf 0.35	0.616	0.582	0.655	317/228/167
Detector, conf 0.50	0.627	0.603	0.653	316/208/168
Sequential step	F1	Δ		TP/FP/FN
Drop missed components	0.759	−0.131		329/54/155
Add spurious detections	0.687	−0.072		322/131/162
Use detected geometry	0.644	−0.043		329/208/155
Use detected classes	0.627	−0.017		316/208/168

G. END-TO-END WITH DETECTED COMPONENTS

We replaced the annotated boxes with the outputs of our trained component detector, a YOLO26m-OBB model with 16 merged classes that reaches a mean average precision (mAP@0.5) of 89.0% on its validation split, with its lowest class recall (70.7%) on crossovers. The detector runs on the CGHD original, corrected for its EXIF orientation tag, at its training size (long side 1024 px), and its boxes are mapped into the benchmark frame. We use the deployment confidence threshold of 0.5 without tuning. On the electrical components, detection precision, recall and F1 are 0.886, 0.858 and 0.872, and the same when the class must also match. As a check of the scoring path, feeding the annotated boxes through the same mapping and matching code reproduces 418/37/66.

Table 9 gives the results. End-to-end micro-F1 is 0.627 [0.519, 0.731] (macro-F1 0.656), a change of -0.263 [-0.366 , -0.170] from annotated boxes; the lower thresholds give 0.613 and 0.616. Replacing the annotated boxes one factor at a time attributes -0.131 to missed electrical components, -0.072 to spurious detections, -0.043 to box localization and -0.017 to classification; with a sequential decomposition, the shares depend on the order of the steps. Missed components cost the most because every pair involving a missed component becomes a false negative. The detector is sensitive to orientation (its training used rotation augmentation of at most $\pm 10^\circ$). On the 14 images whose benchmark copy is re-oriented relative to the CGHD file, end-to-end micro-F1 is 0.491, against 0.739 on the other 17.

H. CROSSOVERS

Edge types 2, 4 and 5 do not consult crossover labels, so a missed crossover could merge two independent wires. The 31 images contain 13 annotated crossovers in eight images. With annotated boxes, recall on these eight images is 0.834, against 0.884 on the 23 without a crossover ($p = 0.51$). In C242_D1_P1, suppressing the two endpoint–endpoint edges adjacent to crossings changes TP/FP/FN from 27/4/0 to 19/0/8: the edges that create the false pairs also carry true connections. Deleting the annotated crossover boxes before joining changes predictions on 6 of the 8 crossover images

TABLE 10. Micro-F1 of our join by CGHD drafter on the 31-image benchmark (drafters with at least four images).

Drafter	Images	F1	F1 _{macro}	TP/FP/FN
drafter_1	7	0.933	0.907	56/6/2
drafter_2	5	0.854	0.874	85/14/15
drafter_10	5	0.987	0.995	37/1/0
drafter_12	4	0.913	0.938	21/0/4
All	31	0.890	0.901	418/37/66

and raises micro-F1 by 0.010, so on annotated boxes the crossover labels do not help the join overall. In the end-to-end run, the detector finds all 13 crossovers and labels each as a crossover at every tested threshold, so none of the false pairs in Table 9 comes from a missed or misclassified crossover; removing the detected crossovers lowers end-to-end micro-F1 by 0.019. These images were probably in the detector's training split, so this is not a held-out estimate of crossover recall, and thirteen instances are too few to test the 70.7% validation recall.

I. VLM REFERENCE

We ran Claude Opus 4.8 as a reference system. It receives the annotated boxes of the electrical components, drawn on the image and numbered without class labels; our join receives the same boxes with their classes, and neither method receives annotated wires. The VLM was run on the human-verified benchmark only. The model returns JSON sets of component terminals that share an electrical node; each image is a separate call with no shared context. We score its output with the same component-pair metric.

The VLM reaches micro-F1 0.923 (P 0.970, R 0.880; macro-F1 0.949) and is exact on 21 of 31 images, against 11 for our join. On pooled pairs, our join trails by -0.033 [-0.078 , $+0.008$], which is not significant (Holm-adjusted bootstrap $p = 0.12$, permutation $p = 0.15$; Table 4). Per image, the VLM is better: it wins 13 images, ties 11 and loses 7 (Wilcoxon $p = 0.030$). Its errors follow the same complexity trend as ours (Table 6). The two systems differ in what they require: the VLM needs a call to a proprietary API for every image and returns no geometric trace of its decisions, whereas our join runs locally and deterministically without connectivity training data, and every recovered connection maps to a wire segment or a completion edge.

J. PER-DRAFTER RESULTS

Table 10 groups the 31-image results by CGHD drafter, using the drafter paths of the published CGHD-1152 file index, which place every benchmark filename under exactly one drafter. The four drafters with at least four images range from micro-F1 0.854 to 0.987. The held-out benchmark gives the spread over all 24 drafters (Section V-D).

VI. DISCUSSION

The endpoint-graph join with completion is the strongest deterministic method on both real benchmarks, and its margins

are significant on both, which differ in labels, images and drafters. The held-out benchmark addresses the three main weaknesses of the human-verified set: its labels come from the dataset authors' annotations and not from a join, it covers 24 of the 25 drafters, and its images were not used to develop the join. Completion accounts for most of the gain over the base graphs, and on the high-precision scale-relative base it adds recall while keeping precision above 0.9.

The experiments also locate the remaining errors. With annotated components on clean copies, extracted wires are as good as annotated ones. On difficult photographs, wire extraction limits accuracy; with detected components, missed detections do. Three changes follow from these results: an extractor whose pixel parameters follow the component scale, resampling inputs to a nominal component size, and a detector trained with full rotation augmentation. The VLM reference shows that a general model given component boxes can exceed a geometric join per image on this task. The geometric join remains useful where a local, deterministic and inspectable result is required.

A. SCOPE AND LIMITATIONS

Task and metric. The pipeline recovers structural connectivity. It does not read component values or device models, so its SPICE export is illustrative and uses placeholder values; simulation of real scans is not evaluated. The component-pair metric ignores terminal identity: a perfect score does not certify pin assignments, the exact net partition or the absence of shorts between pins of one component. The completion guard blocks new same-component merges but does not repair merges made by the base graph. Connectivity is read from drawn wires only: terminals and supply symbols are wire ends, so identically labeled supply terminals (e.g., two "+5 V" points) and the implied source between a pair of input terminals are not interpreted, which would require reading their labels. Table 11 lists the device groups: pin templates for transformers, optocouplers and other complex devices exist but are not validated, and the detector merges several original classes into one label.

Data. All images come from one corpus. The human-verified nets were bootstrapped from a conservative endpoint-graph join and corrected by a single annotator, which may leave a bias toward graph-based methods; on the held-out benchmark, with labels from the dataset's own annotations, our join again ranks first. The inspection of the 17 overlaps found five human-label errors (four in C242_D1_P1); we left the released human-verified labels unchanged. The derived reference is not human-verified, mainly misses connections, and its quality filter and a junction rule were fixed after inspecting the 17 overlapping images. The benchmark copies of the 31 images are resized, partly re-oriented, and half stroke maps, which makes them easier than raw photographs of the same drawings. The Hough and CCL baselines keep their 31-image settings on the held-out photographs.

Scale. The join tolerances derive from one median scale per image, and their clamps are tuned to native resolution. A

component much smaller than the median can create a spurious pair, as in the synthetic Wheatstone case. The extractor's pixel-valued parameters limit accuracy on downscaled inputs. Rescaling tests changes in zoom, not in drawing style, and uses annotated components.

Detector. The detector was trained on CGHD images with an 85/15 random split, and no manifest of that split survives. The 31 benchmark drawings come from drafters eligible for training, and about 26 of them would be expected in the training split, so the end-to-end numbers are likely optimistic for detection. The detector excludes drafter_0, whose drawing style differs from the rest of the corpus, and it is sensitive to orientation.

Coverage. The human-verified benchmark contains 13 annotated crossovers, and we did not evaluate dense multi-track buses or multi-layer crossings. The VLM reference uses one model and one prompt, and was run only on the human-verified benchmark. The synthetic error model is authored and not calibrated to measured detector errors.

Expansion. The derived reference makes a larger human-verified benchmark affordable: its nets can seed the verification tool, which logs a timestamp at every save so that per-image effort can be measured. We plan to sample new images stratified by component count, drafter, crossing and bus structure and capture conditions, have two annotators correct each image independently with adjudication of disagreements, and extend the evaluation to printed schematics and to corpora other than CGHD.

VII. CONCLUSION

We presented a deterministic pipeline for structural netlist recovery from hand-drawn schematics, built on an occlusion-first wire extractor, a typed endpoint-graph join and degree-budget completion. With annotated components, the join reaches component-pair micro-F1 0.890 on 31 human-verified images and 0.711 on 164 held-out photographs with independently derived reference nets, and it beats every deterministic baseline on both with significant paired margins. Its accuracy holds from $0.5\times$ to $3\times$ image scale when wires are given, and it reaches 0.627 end to end with a trained detector. A VLM given the same components scores higher per image, at the cost of a proprietary call per image. Better extraction on difficult photographs and better component detection are the most direct routes to higher end-to-end accuracy.

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TABLE 11. Connectivity evaluation, pin construction and export by device group.

Device group	Evaluated scope	Pin construction	Export
R/C/L/D/Q, voltage sources and ICs	Component-pair connectivity on both benchmarks	Geometric templates; endpoint clustering for discrete and source types, generic layouts for ICs	Placeholder values; ICs are skipped because no export model is implemented.
Switches	Scored in the extended held-out variant (Section V-D); not in the primary subset	Two pins at the shorter OBB-edge midpoints; long-axis fallback	A 0.001 Ω resistor if the two pins map to distinct nodes; no switch state is recovered.
Transformers, thyristors, optocouplers and other complex devices	Not evaluated	Two-pin geometric guess for transformers; optocoupler template; generic fallback	No export model; skipped.

DATA AND CODE AVAILABILITY

The pipeline, the synthetic-circuit generator, the 31-image human-verified net-level ground truth (ground_truth/real_net), the net-level verification tool used to produce it, the CGHD-derived reference nets of the held-out benchmark (ground_truth/cghd_ref/), the join-strategy and baseline benchmarks, the Claude-VLM connectivity harness and the evaluation scripts for the paired tests, strata, rescaling, held-out and end-to-end experiments (wire_detection/benchmark/revision2/) are available at <https://github.com/boscochanam/circuit-digitization>, together with the image-to-drafter mapping used in Table 10. The reference nets are derived from the CGHD annotations [14]; no CGHD images are redistributed. The component-detection model is available at <https://huggingface.co/boscochanam/circuit-component-detector>, trained on CGHD images [14].

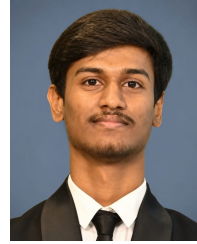
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Fuzzy Logic and Intelligent Systems.

BOSCO CHANAM received the B.Tech. degree in Computer Science and Engineering from the Symbiosis Institute of Technology (SIT), Pune, India. He is currently an AI/ML Engineer at the AI Incubation Centre at Bharat Forge (Kalyani Group), Pune, India. His professional and research interests include computer vision, edge AI optimization, and robotics. Mr. Chanam has co-authored research in natural language processing benchmarking, published in the *International Journal of*



ASHAY KUMAR SINGH received the B.Tech. degree in Computer Science and Engineering from the Symbiosis Institute of Technology (SIT), Pune, India. His research interests lie in machine learning, computer vision, Retrieval-Augmented Generation (RAG), optical character recognition (OCR), and multimodal AI.



CHRIS DCOSTA received the B.Tech. degree in Computer Science and Engineering with Honors in Artificial Intelligence and Machine Learning from the Symbiosis Institute of Technology (SIT), Pune, India. His research interests include computer vision, machine learning, natural language processing, and deep learning.



ARGHADEEP DAS received the B.Tech. degree in Computer Science and Engineering from the Symbiosis Institute of Technology (SIT), Pune, India. His research interests include reinforcement learning, world models, computer vision, 3D rendering, agentic AI, embodied AI, robotics, and machine learning.

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PRANAVESH KUMAR TALUPURI is with the Department of Computer Science, University of Southern California, Los Angeles, CA, USA. His research interests include machine learning, deep learning, and computer vision.



SHWETAMBARI CHIWHANE holds a B.E. in Computer Science and Engineering from Sant Gadge Baba Amravati University, Amravati, India, and an M.Tech. in Computer Science and Engineering from Dr. Babasaheb Technological University, Lonere, India. She earned her PhD in Computer Engineering specializing from Bharath Institute of Higher Education and Research, Chennai, India. Presently, she serves as an Assistant Professor in the Department of Computer Science and

Engineering at Symbiosis Institute of Technology, Symbiosis International (Deemed University), Pune, India, gathering over two decades of teaching and research expertise. She is life member of ISTE.

Her contributions extend beyond academia; she has contributed as a reviewer for various international conferences. Dr. Shwetambari's research expertise is evident with her receiving many publications. She has authored over 35 research articles in various international and national conferences and journals. Her professional engagements include roles such as NAAC, NBA, among others.

Recognized as a PhD guide in Symbiosis International (Deemed University), Presently she is guiding 4 research scholar and she has guided more than 100 undergraduate students for their research endeavors. Her current research interests revolve around Image Processing, Computer Vision, Artificial Intelligence and Machine Learning.